UNSW MARKETING ANALYTICS HACKATHON 2026

# Getting slow loans funded faster

What 1.45 million Kiva loans say about predicting slow loans and supporting them

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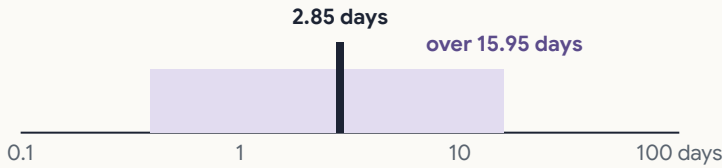


# Act on predictable delays now; test writing advice before giving it

Problem, solution and proof across all evidence layers

## THE PROBLEM

**2.85 days**

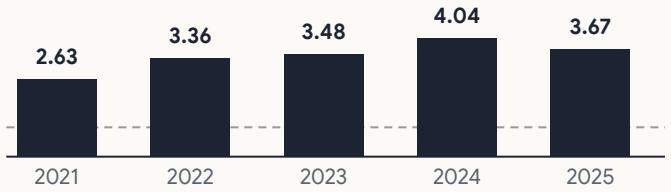



### Median wait to full funding

A quarter of loans wait over 15.95 days.

## OUR SOLUTION

**2.6–4.0×**

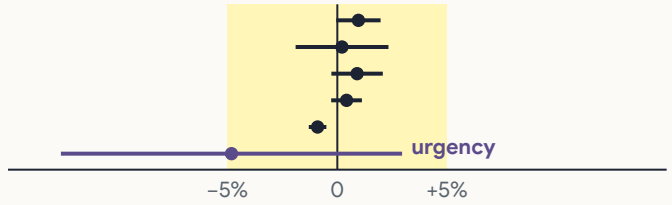



### Better than chance at flagging slow loans

Loan details flag the slowest fifth early, so support can go there first.

## THE PROOF

**±5%**




### Framing links stay small

Five framings stay within ±5% of funding time, so rewriting stories is not the lever.

**Our ask to Kiva:** pilot a slow-funding flag with lender-side support, and test one truthful prompt.

Predict  Prioritise  Test

For whom and when: no period, loan size, gender or region shows a large framing effect; a few rare-word patterns are hypotheses to test.

# Agenda

01 → 02 → 03 → 04

## The problem

Who waits, and where?


How a loan gets funded 4
The wait 5
Where loans and delays sit 6


## What drives the wait

What does the evidence show?


What we read in each loan 7
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For whom and when 13
What text adds 14
Fragile across countries 15


## Our solution

What can Kiva do now?


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Predict 17
Prioritise 18
Test 19

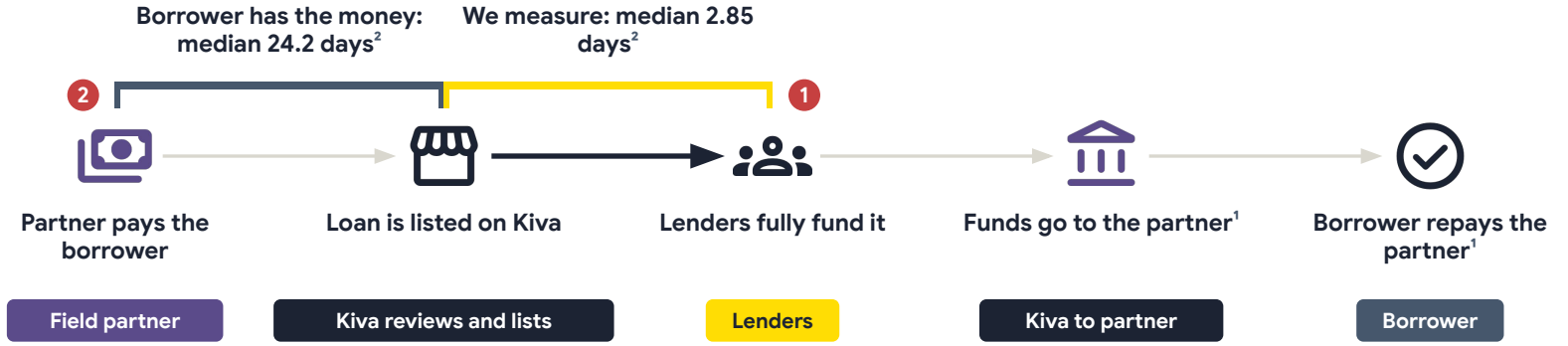

## Safe next steps

What keeps it honest, and what comes next?


Guardrails 20
Roadmap 21

# Borrowers usually have the money before lenders fund the loan

Descriptive: how a typical pre-disbursed Kiva loan moves, who acts at each stage, and what we measure




### Loans disbursed to the borrower before listing




### Who is in the data

**In the data:** 1,453,840 completed loans, including 1,637 refunded

**Not in the data:** listings that expired or never funded 4

**Figure 1.** Journey of a typical pre-disbursed Kiva loan, not to scale; medians from the full-data audit.

[1] Typical field-partner flow; these steps are not observed in the data.

[2] Separate medians: 24.2 days covers the 96.4% disbursed before listing, and the two do not add up along one loan.

**Scope:** 1,453,846 input records; 6 negative durations excluded; 1,637 refunded records included.

### 1 We study the lender-side step

Our outcome runs from **listing to full funding**, median 2.85 days.

### 2 Borrowers usually already have the money

**96.4% of loans** reach the borrower a median **24.2 days** before listing.

### 3 Speed turns over partner capital

Only **3.6%** are paid out after listing, so slow funding mainly holds up the partner's next loans.

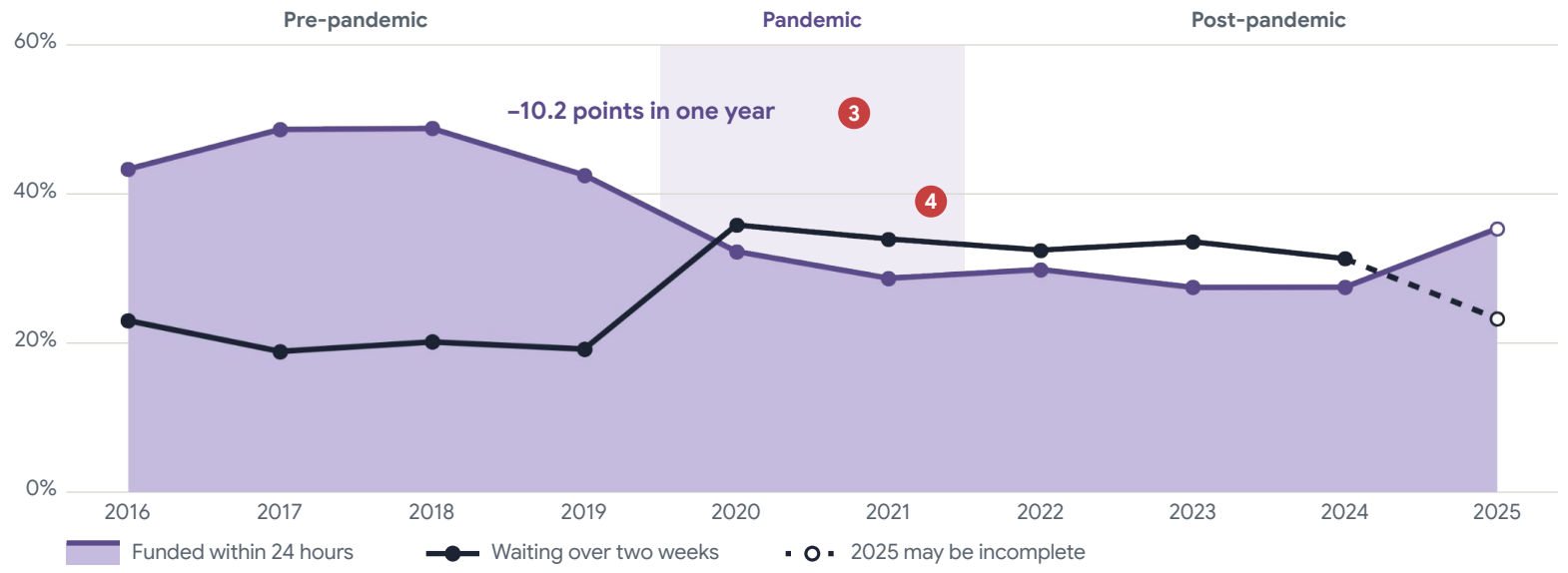
### 4 Unfunded loans are missing

Only loans that **eventually funded** are in the data; expired listings are not.

→ Why speed is a lender-side problem

# Waits got longer after 2019: a third of loans now wait over two weeks

Descriptive: platform funding time from listing to full funding, 1,453,840 completed loans, by posting year



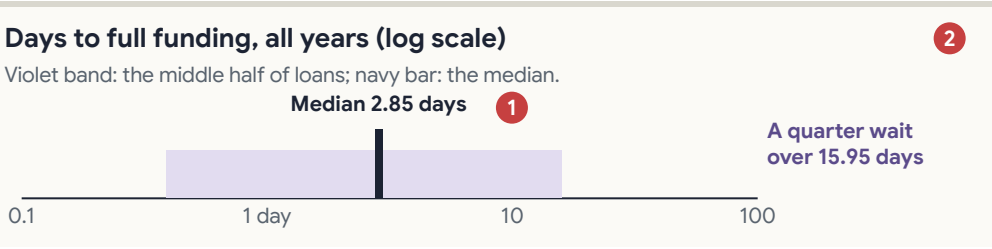






**Figure 2.** Same-day share and share waiting over two weeks by posting year; funding-time quartiles, all years.

### 1 Most loans fund within days

Half are fully funded within **2.85 days**, a quarter within **0.39 days**.

### 2 A long tail waits weeks

A quarter of loans wait **over 15.95 days**, while borrowers already have the money.

### 3 Same-day funding fell in one year

From **42.5% in 2019 to 32.3% in 2020**, a fall of 10.2 points.

### 4 Long waits became common

Loans waiting over two weeks rose from **19.2% to 35.8%**; in 2021 that was **55,408 loans**.

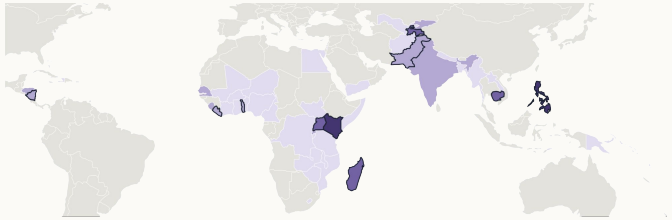
→ Why early warning matters

# Over half of all two-week waits come from just three countries

Descriptive: where loans sit and where they wait over two weeks, by country, 1,453,840 completed loans

### (a) Where the loans are

Ten of 48 countries hold 81% of loans (outlined on the map); table share cells use the map's colour scale.




| | Country | Region | Loans | Share | Same-day |
|---|---|---|---|---|---|
| 1 | **Philippines** | Asia | 521,586 | 35.9% | 48% |
| 2 | **Kenya** | Africa | 266,562 | 18.3% | 29% |
| 3 | **Uganda** | Africa | 69,651 | 4.8% | 31% |
| 4 | **Tajikistan** | Asia | 68,865 | 4.7% | 18% |
| 5 | **Cambodia** | Asia | 58,074 | 4.0% | 49% |
| 6 | **Madagascar** | Africa | 55,670 | 3.8% | 67% |
| 7 | **Pakistan** | Asia | 42,767 | 2.9% | 15% |
| 8 | **Nicaragua** | Central America | 38,520 | 2.6% | 15% |
| 9 | **Liberia** | Africa | 28,847 | 2.0% | 28% |
| 10 | **Togo** | Africa | 27,513 | 1.9% | 38% |

### (b) Where the long waits are

Each bar is all of a country's loans, split by how long they took to fund.

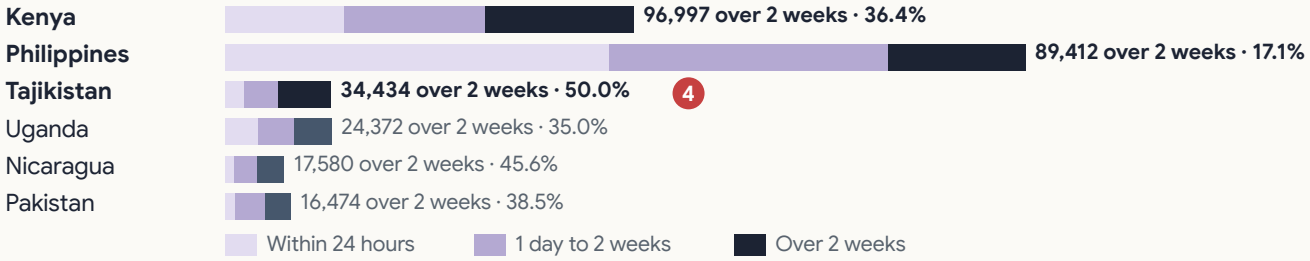



**Figure 3.** Share of all loans by country (a, map and table) and loans waiting over two weeks (b); same-day: funded within 24 hours.

### 1 Ten countries hold most loans

Ten of 48 countries hold **81% of loans**; Asia and Africa hold **92.8%**.

### 2 Speed varies widely by country

Same-day funding runs from **15% in Pakistan to 67% in Madagascar**.

### 3 Three countries hold half the long waits

Kenya, the Philippines and Tajikistan hold **56.1%** of the **393,847** loans that waited over two weeks.

### 4 Large markets wait too

In Kenya **36.4%** of loans wait over two weeks; in Tajikistan **50.0%**.

→ Why the flag must work country by country

# Each loan story is scored on eight framing measures

Illustrative listing: what is fixed when a loan is listed, and what we measure in the description



**FIXED AT LISTING** 1

| Amount, median | Term, median | Borrowers, median |
|---|---|---|
| **US$325** | **12 months** | **1** |
| Sector, most common | Country, most common | Partner |
| **Agriculture** | **Philippines** | **Field partner** |

**EDITABLE DESCRIPTION** 4

> She has a banana stall at the market to support her family. She wants to buy more stock quickly, before prices rise. She is thankful to every lender who helps her business grow.

**The eight framing measures, counted per 100 words, with the share of loans using each (plus a sentiment score)** 2

- **Family** — Relatives and schooling. e.g. children · in 88.7%
- **Business** — Trade and stock words. e.g. market · in 85.0%
- **Basic needs** — Water, health, housing. e.g. water · in 23.3%
- **Agency** — Planning and self-direction. e.g. decided · in 54.1%
- **Gratitude** — Thanks and blessings. e.g. thankful · in 19.6%
- **Urgency** — Time-pressure words. e.g. quickly · in 1.3%
- **First person** — The borrower's own voice. e.g. my · in 1.0%
- **Third person** — Written about the borrower. e.g. her · in 97.3%

3

**Figure 4.** Illustrative composite listing, not a real borrower; dataset medians and most common categories. Word lists are not yet validated (gate 1).

### 1 Most of the listing is fixed

Amount, term, sector, country and partner are set; the **description** is the part a partner writes.

### 2 We score eight framing measures

Counted with **word lists per 100 words**, plus VADER sentiment and text form.

### 3 Some measures are rare

Urgency appears in **1.3%** of descriptions and first person in **1.0%**; only **11.3%** never mention family.

### 4 Pronouns mostly reflect the writer

Descriptions are **written about the borrower**, so third person is context.

→ Why wording is tested, not assumed

# Slow funding is a lender-side delay that loan details predict, not wording

Three findings across the outcome audit, the early-warning check and the equivalence test, each leading to a step of the solution

### The wait is on the lender side

1

**96.4%**

of loans reach the borrower before listing




→4

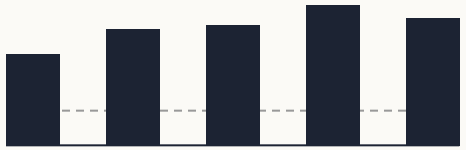
Leads to **Prioritise**

### Loan details predict who waits

2

**2.6–4.0×**

better than chance at flagging the slowest fifth




→17

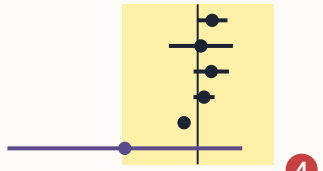
Leads to **Predict**

### Wording makes little difference

3

**±5%**

five framings stay within this share of funding time




→12 →13

Leads to **Test first**

## 1 Act on the lender side

**96.4%** of loans reach the borrower before listing, so the wait ties up partner capital, not the borrower.

## 2 Predict slow loans early

A boosted tree on loan structure flags the slowest fifth **2.6–4.0× better than chance**.

## 3 Don't chase framing tweaks

Five framings stay **within ±5%** of funding time, and text adds little to the flag.

## 4 It holds in every group

Six framings stay small in **all 10 subgroups** of period, loan size, gender and region.

→ The evidence behind each insight follows

# Four analyses feed the solution; none is a causal claim

One dataset: the equivalence test builds on the robustness checks, and the forecast benchmark asks a separate question

**Kiva loans, 2016–2025: 1,453,840 valid completed durations in 48 countries** 1

### Explanatory models

**Which features are linked to funding speed?**



64 of 120 significant terms lost significance

Feeds Test 2

### Robustness checks

**Do the links survive richer controls?**

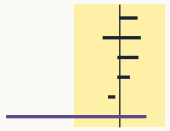



No signal survives a stricter test

Feeds Prioritise 3

### Equivalence test

**Which links are small enough to rule out a large effect?**



Five framings small; urgency unresolved

Feeds Prioritise

### Forecast benchmark

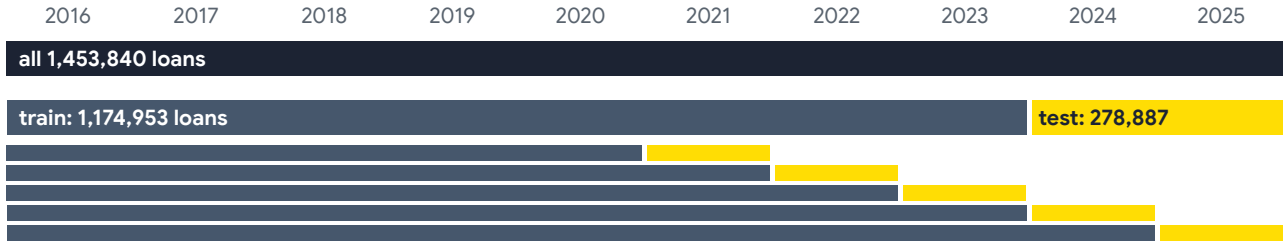
**Does text improve a forecast built on loan structure?**



Small gain: 0.26–0.39 days, about half context

Feeds Predict 4

**Years each analysis uses**




## 1 One dataset, four questions

All four use the same **1,453,840 loans**, and each supports a different kind of claim.

## 2 Uncertainty comes first

Clustering by 48 countries removed **64 of 120** significant terms.

## 3 Framing signals stay small

After a wild cluster bootstrap, no term passes: the lowest **p is 0.136**.

## 4 Forecasting is tested separately

Each model trains only on **loans completed before its test year**.

→ None is causal, so the solution ends with a test

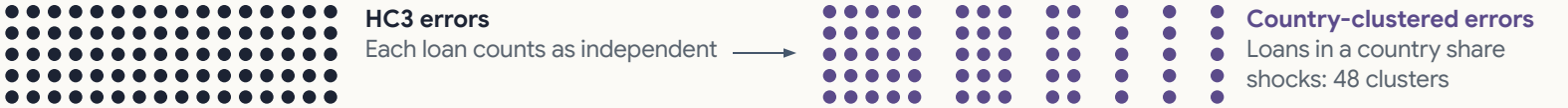
# Clustering by country removed 64 of 120 significant terms

Explanatory models: HC3 versus 48-country clustered errors on 1,453,840 loans

### Why the errors widen

1

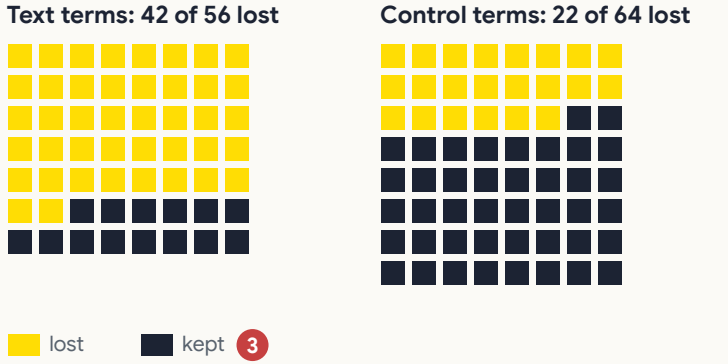
The loans and estimates stay the same; only the assumption about how loans vary together changes.




### Which terms lost significance

2

Each square is one term significant under HC3.




### Urgency before and after

4

p-value for urgency in each model; clustering moves both past the 0.05 line.

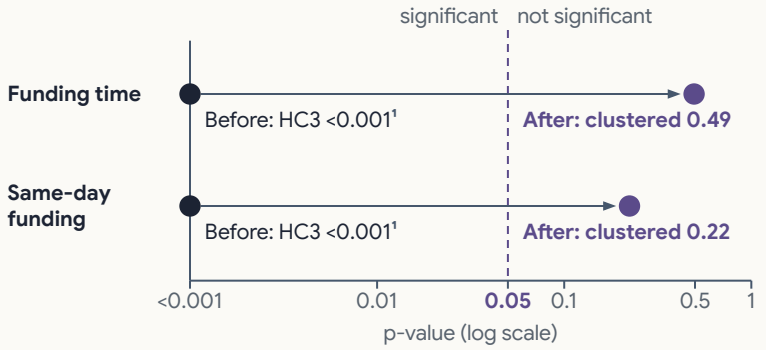



**Figure 5.** Explanatory models, HC3 versus errors clustered by country (48 clusters). Dots in the top panel are illustrative.

[1] Printed as 0.0000 in the saved run. Text terms include interactions of a text measure with a control.

### 1 Few countries dominate

Ten countries hold **81% of loans**; regional patterns rest on **two countries each**.

### 2 Text terms are hit hardest

**42 of 56 text terms** lost significance, against 22 of 64 controls.

### 3 Few text terms survive

Only **14** stay significant: sentiment in both models and 12 family-framing interactions.

### 4 Urgency no longer passes

Its p-value rose from below 0.001 to **0.49** for funding time and 0.22 for same-day funding.

→ Why the flag is checked country by country

# No framing signal survives stricter tests, and what remains is tiny

Robustness checks: programme controls, then a wild cluster bootstrap across 48 countries

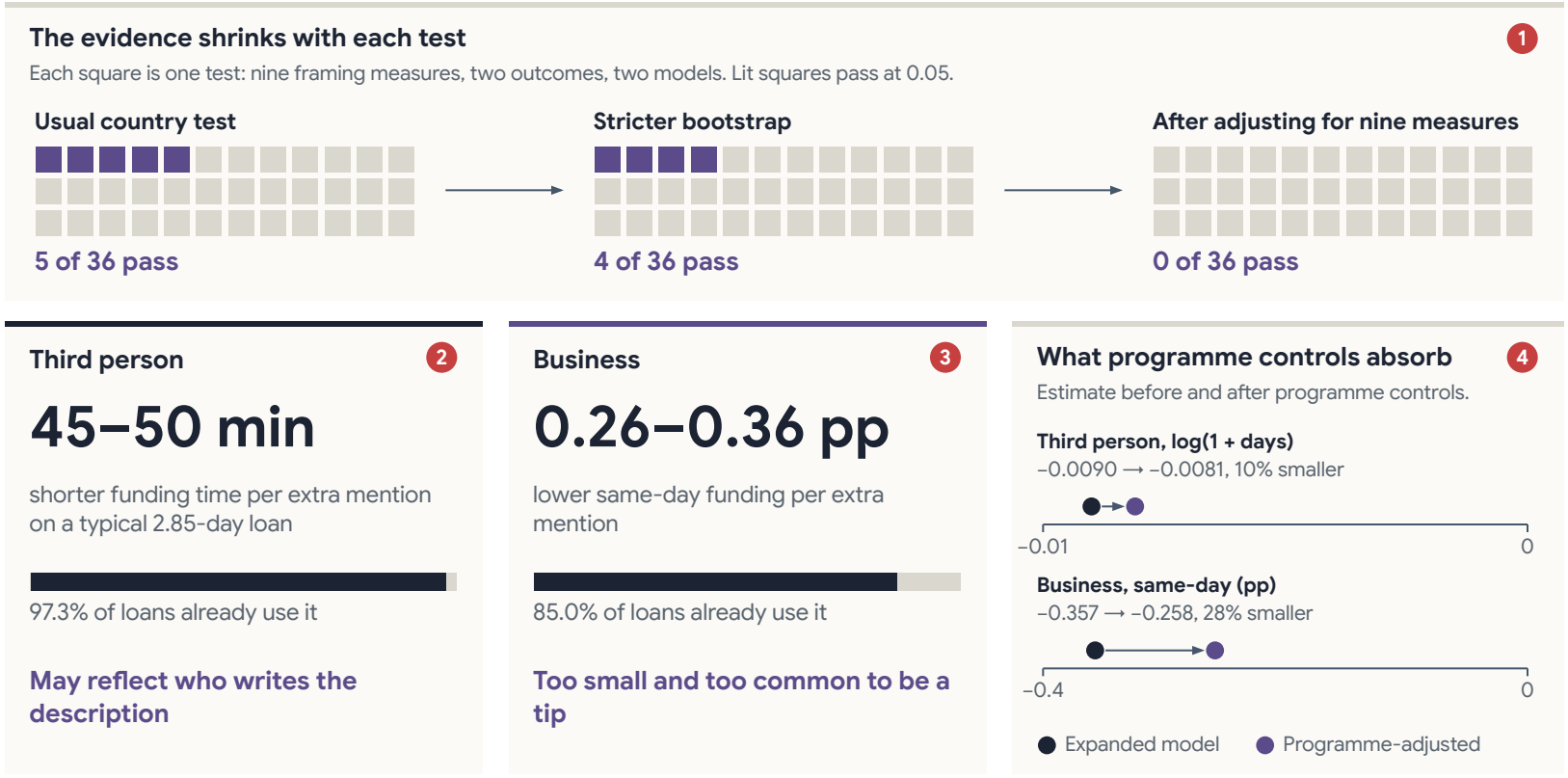



**Figure 6.** Per extra mention per 100 words; BH-adjusted wild cluster bootstrap p across nine terms (Webb weights, 9,999 draws). Typical loan: the 2.85-day median.

### 1 The evidence shrinks with each test

**5 of 36** tests pass the usual country test, 4 the bootstrap, **none** after adjustment (lowest p 0.136).

### 2 What remains is tiny

**45–50 minutes** shorter funding per third-person mention; **0.26–0.36 pp** lower same-day per business mention.

### 3 These words are already common

Third person is in **97.3%** of loans, business in 85.0%: no ready writing tip.

### 4 Pronouns may reflect the writer

Programme controls shrink the links **10%** and **28%**; partners usually write the text.

→ Why we prioritise loans, not wording

# Five framings have small links to funding time; urgency remains unknown

Equivalence test of the robustness-check coefficients; exploratory, bound set after the estimates

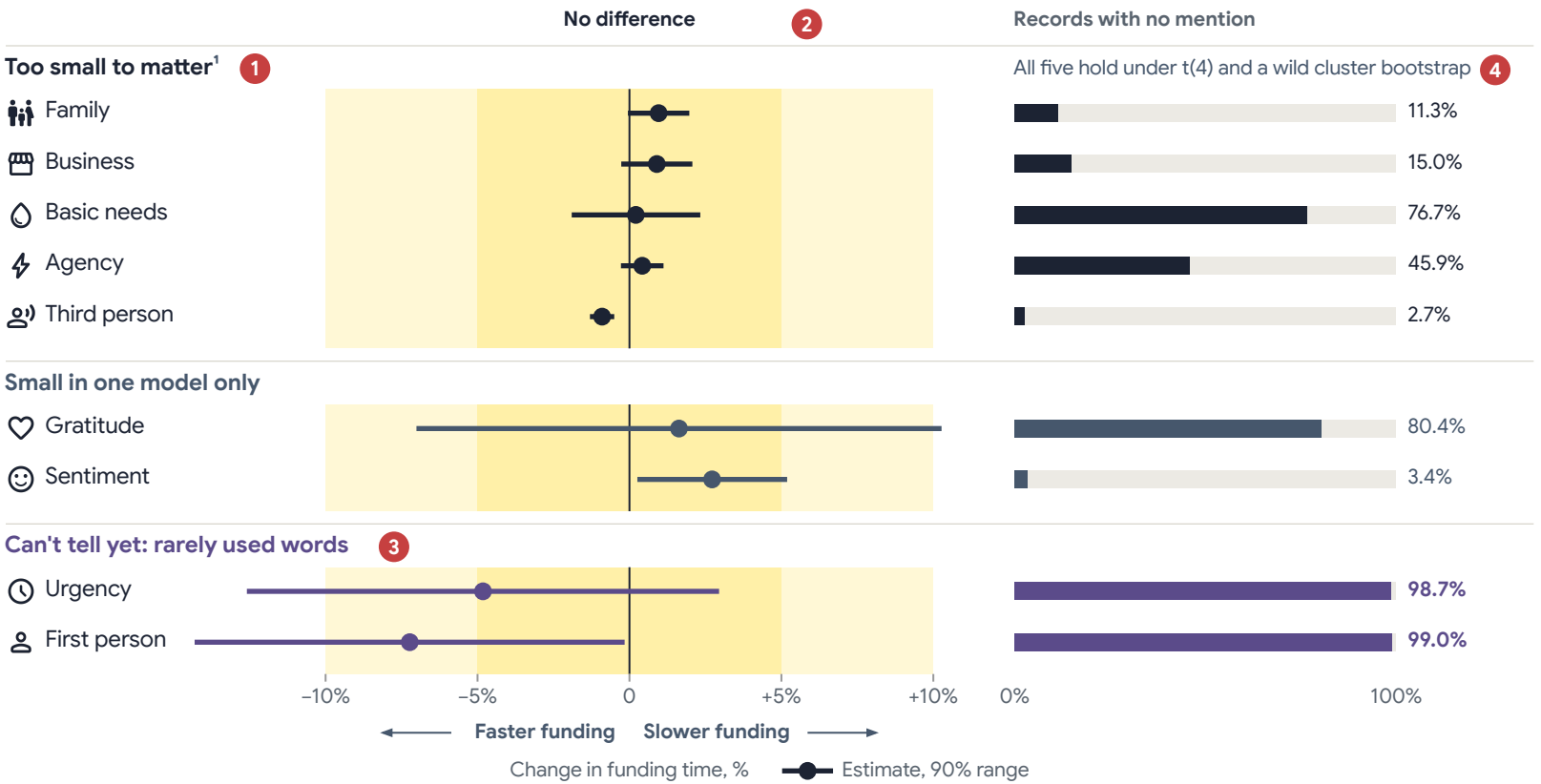



**Figure 7.** Change in funding time per extra mention per 100 words (sentiment: per +0.5 score), expanded model, 1,453,840 loans.

[1] Within ±5% of funding time: ±0.05 on log(1 + days), BH-adjusted; bound set after the estimates.

### 1 Five framings have small links

Family, business, basic needs, agency and third person stay **within ±5%** in both models.

### 2 Small means under five hours

±5% is about **4.5–4.7 hours** on a typical 2.85-day loan.

### 3 Urgency is too rare to judge

Under **2% of descriptions** use urgency or first person; their ranges still reach **6–14%**.

### 4 The verdicts hold under stricter tests

A wild cluster bootstrap keeps **35 of 36 verdicts**; three framings stay small even within **±2%**.

→ Why we don't chase framing tweaks

# Six framings stay small in every period, loan size, gender and region

Exploratory: programme-adjusted model refitted in 10 subgroups; ±5% equivalence test, bound set after the estimates

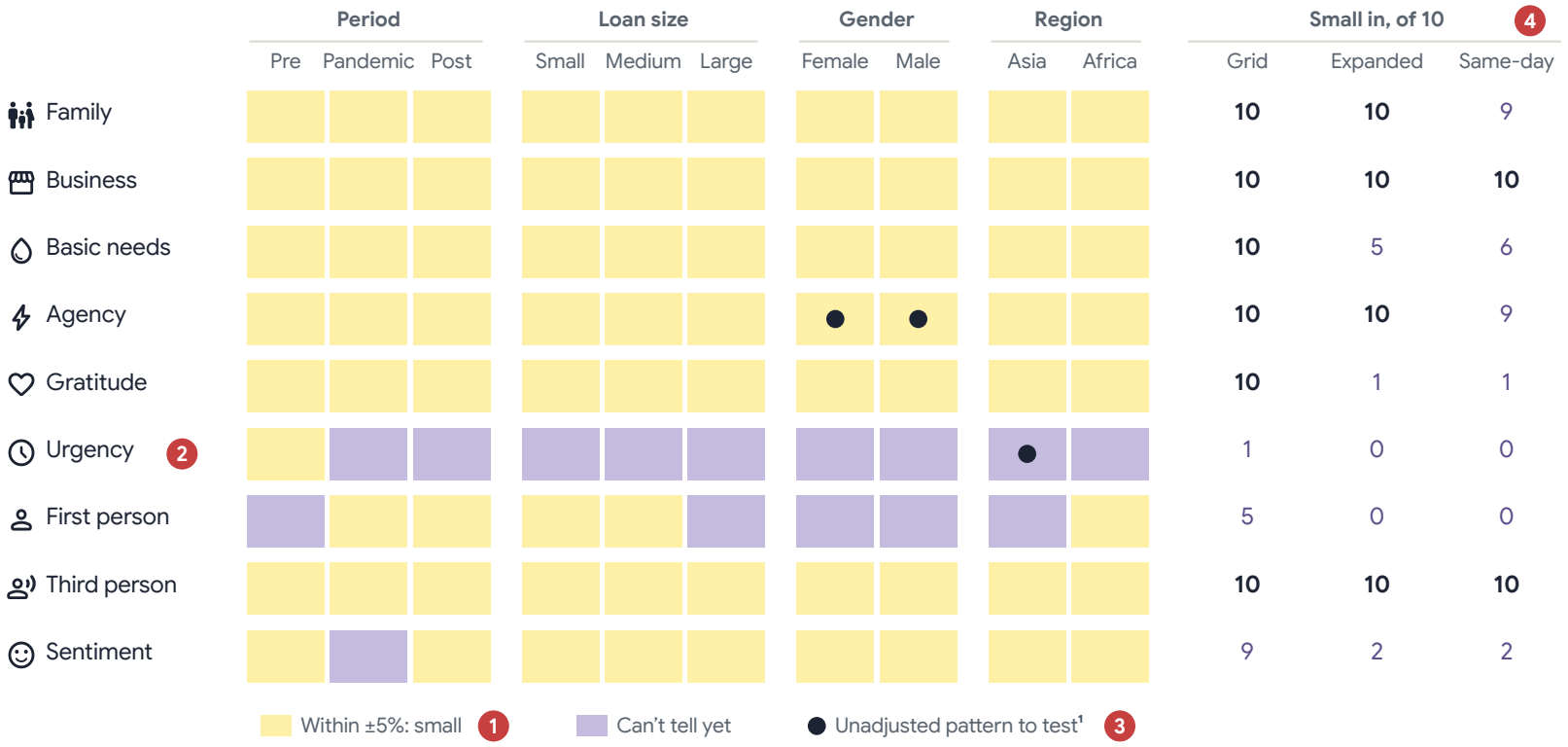



**Figure 8.** Funding-time verdict by framing and subgroup, programme-adjusted model, 1,453,840 loans. Expanded: funding time. Same-day: ±1 pp.

[1] Unadjusted p-values among about 180 subgroup tests; hypotheses for the Test step, not findings.

### 1 Small everywhere, not just on average

The five framings on the last slide plus gratitude stay within ±5% in **all 10 subgroups**; no family interaction with period or loan size survives clustering.

### 2 Rare words stay open in most groups

Urgency is small only before the pandemic (**1 of 10**); first person in **5 of 10**.

### 3 Two patterns to test, not to act on

Urgency in Asia and agency by gender stand out among about **180** unadjusted tests.

### 4 The count depends on the model

In the expanded model **4** framings stay small in all 10; for same-day funding, **2**.

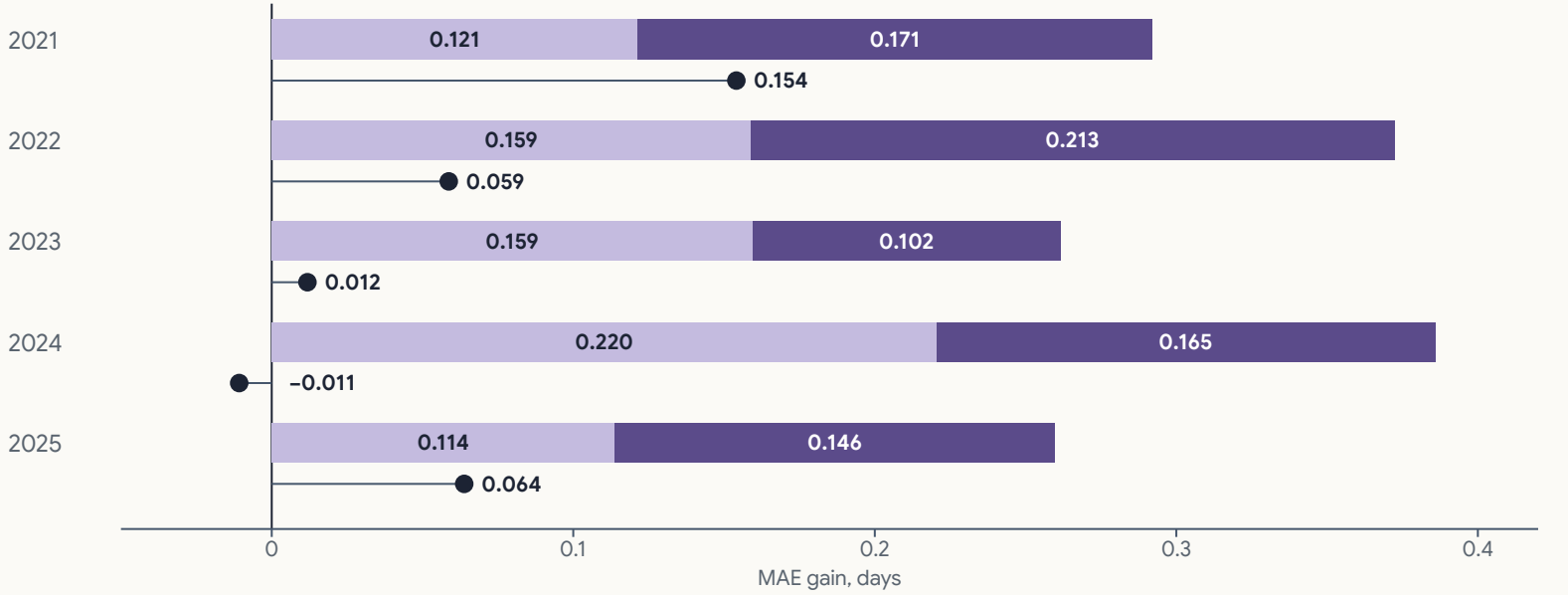
→ Why one playbook serves every group

# Text adds a small, inconsistent forecast gain; about half is context

Forecast benchmark: gain from adding text to loan details, five annual test years, identical rows

**Where the forecast gain comes from** 1

MAE gain in days from adding text to loan details. Bars: linear model. Dots: larger boosted tree. Right of 0 means lower error.




**Text alone is 0.146–1.032 days worse than loan details alone** 4

**Figure 9.** Annual MAE gain in days, identical rows; bars from the placebo refit, within 0.003 days of the benchmark. Text = form, word lists, VADER and TF-IDF.

### 1 A small gain on a simple model

Text, mainly **TF-IDF word patterns**, lowers a linear model's error by 0.26–0.39 days, about 7 hours.

### 2 About half is context

Shuffled text keeps **42–61%** of the gain; the loan-specific part is 0.10–0.21 days.

### 3 A stronger tree gains less

A larger boosted tree gains **at most about 0.15 days**, and under 0.07 in four of five years.

### 4 It does not show rewriting helps

Text alone does worse than loan details, and no forecast tests an edit.

→ Why we predict with loan details

# The text gain is fragile across countries and on a stronger model

Forecast benchmark: annual MAE gain in days, with 95% ranges from resampling months or countries

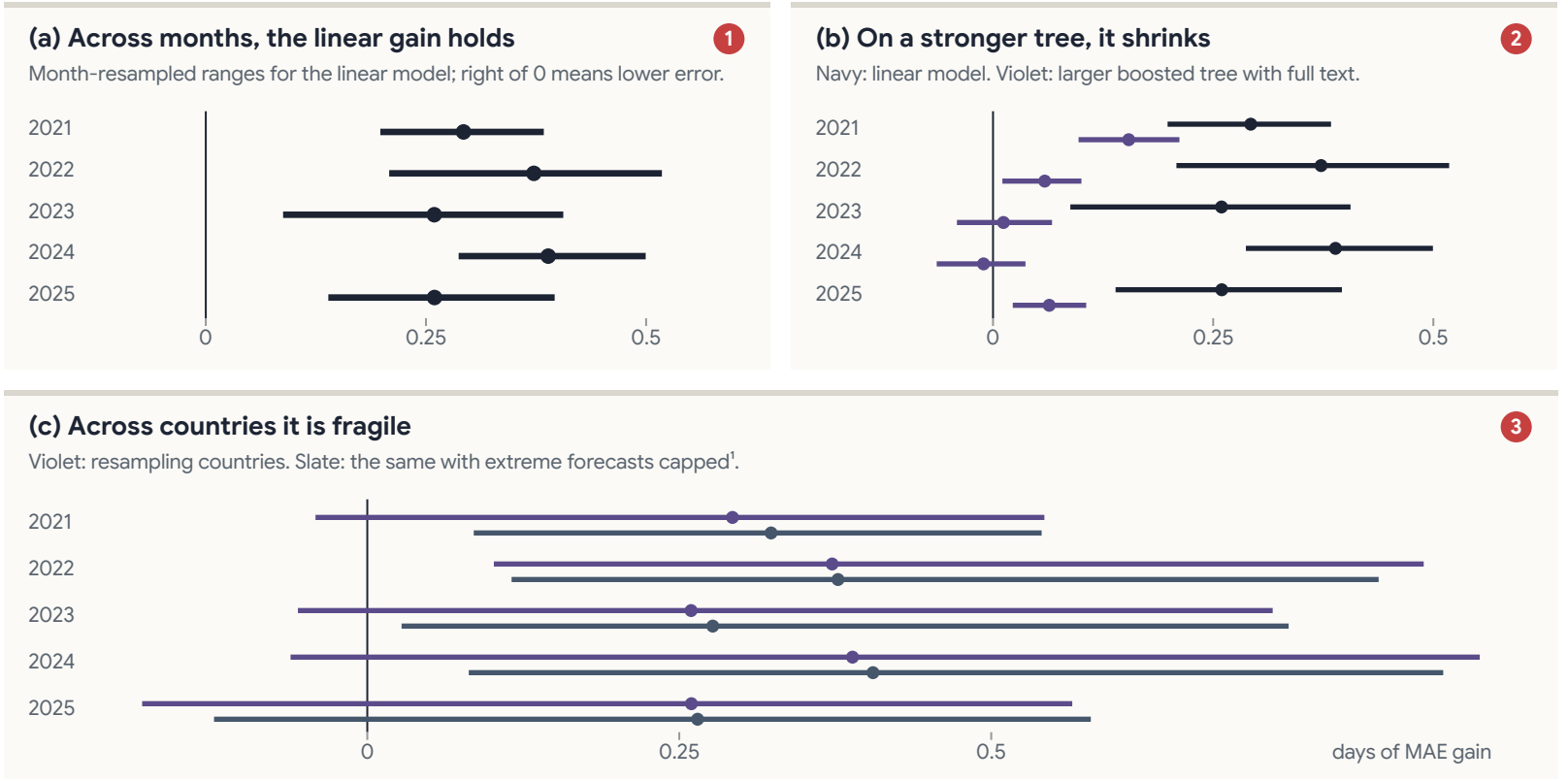



**Figure 10.** Annual MAE gain in days from adding text, identical rows. Text = form, word-list counts, VADER sentiment and TF-IDF word patterns.

[1] Forecasts capped at the longest training wait (173.4 days); a diagnostic, not the reported model.

### 1 Stable across months, linear only

Month ranges stay **above zero in 5 of 5 years**; the median gain is about **7 hours**.

### 2 A stronger tree gains little

Its ranges exclude zero in **3 of 5 years** by month and **1 of 5** by country.

### 3 Across countries it is fragile

Country resampling **includes zero in 4 of 5 years**; the loan-specific part clears zero in 2 of 5.

→ Why the flag needs validating by country

# Our solution: predict slow loans, prioritise support, test before advising

A three-step framework for Kiva and partners, built only on what the evidence supports

01

## Predict

**What**

**Score each listing's risk of slow funding when it is posted.**

**How**

A boosted tree on loan structure; text adds little to this score.

**Evidence**

Flags the slowest fifth of loans 2.6–4.0× better than chance.

02

## Prioritise

**What**

**Send lender-side support to flagged loans.**

**How**

Featuring, matching funds, lender alerts and partner follow-up.

**Evidence**

Framing links stay within ±5%, so don't chase framing tweaks.

03

## Test

**What**

**Earn the right to give writing advice.**

**How**

Pass three gates before any advice.

**Evidence**

No edit to a description has been tested yet.

**Writing advice waits until a tested pilot shows it helps.**

# A slow-funding flag at listing could direct support where it is needed

Concept to pilot: an early-warning card built on the forecast benchmark's retrospective predictions

Listing at posting time

Concept to pilot

**Agriculture · US$325 · 12-month term**

Illustrative listing, not a real borrower

Predicted risk of slow funding

top 10%

**Flag: likely slow to fund**

Chance of same-day funding: low

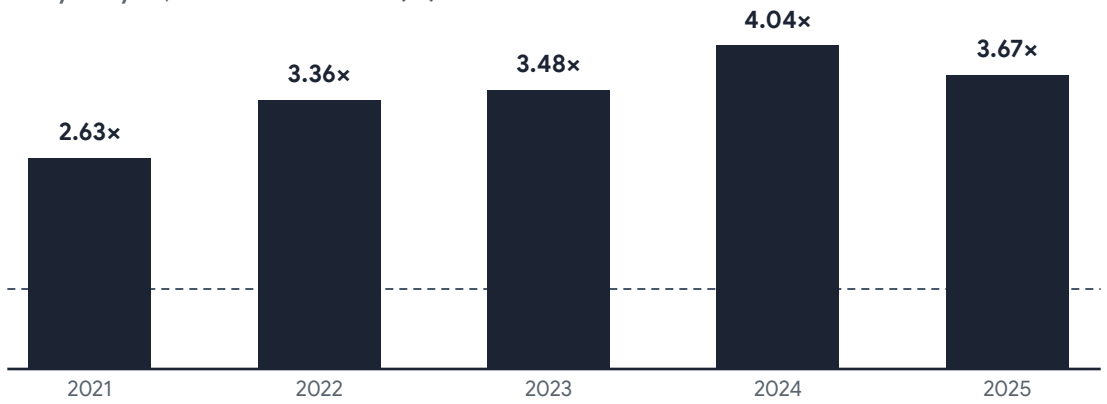
**Suggested support**

- Feature it to lenders
- Offer matching funds
- Alert lenders who follow the sector
- Ask the partner to review it

**2.6–4.0×**

better than chance at catching the slowest fifth of loans, flagging the top 10% of predicted-slow loans

Lift by test year; dashed line = random (1×)




**Text adds little: the tree's precision changes by −0.003 to +0.005.**

**Uneven across countries: within the 10 largest, precision ranges 0.18–0.90.**

The primary 24-hour model ranks same-day funding with ROC AUC 0.8997 on later loans.

**Figure 11.** Lift when flagging the top 10% of predicted-slow loans, against the slowest 20% of each test year; boosted tree, loan structure only.

**Scope:** tested only on loans that eventually funded; a retrospective evaluation, not a deployment test; a flag shows a loan is likely slow, not why. Probabilities need recalibration before use as thresholds.

# Send lender-side support to flagged loans, not story edits

Proposed levers for Kiva and partners; their effect on speed is to be tested, not yet measured

### Feature it

Show flagged loans to lenders early on the lending page.

### Match it

Offer matching funds where a loan is likely to stall.

### Alert lenders

Notify lenders who follow the loan's sector or country.

### Follow up

Ask the field partner to check the listing is complete and clear.

### Don't chase framing tweaks

**±5%**

Five framings stay within ±5% of funding time, about 4.5–4.7 hours on a typical loan, and text adds little to the flag.





**Scope: the levers are proposals; our data do not measure their effect, so each needs a controlled pilot.**

# Test before advising: three gates before any writing advice

Proposed next research for Kiva and participating partners; each gate must pass before the next

01

## Validate the measures

**Question**

**Do our word counts capture clarity, need and agency?**

**Do**

Build a small codebook; two people code the same narratives and resolve disagreements.

**Done when**

Definitions are frozen before later periods are checked.

02

## Close the data gaps

**Question**

**What did lenders see, and which listings never funded?**

**Do**

Get original texts, partner identifiers, exposure data and expired listings.

**Done when**

Unfunded listings and listing-time text are in the data.

03

## Pilot one prompt

**Question**

**Does editing a description change funding speed?**

**Do**

With partners, test one truthful clarity prompt against a control; urgency in Asia is a candidate.

**Done when**

Threshold, power and guardrails are agreed in advance.

### Each gate unlocks a stronger claim

**Today**
Associations and forecast information

→

**After gate 1**
Measures that mean what we say

→

**After gate 2**
Results for every listing, not only funded ones

→

**After gate 3**
A tested effect of editing

**If a gate fails, stop there: report the uncertainty and give no writing advice.**

# Four guardrails keep the solution honest

What this evidence cannot show, and what to do about it

### Associations, not causes

Limit

## 0 tested edits

Every result is observational; no edit or lever has been tested.

**Pilot each lever and prompt against a control.**

### Funded loans only

Limit

## Unfunded: missing

Listings that expired unfunded are not in the data, so reach is unknown.

**Add expired listings before judging reach.**

### Uneven across countries

Check

## 0.18–0.90

Flag precision within the 10 largest countries; probabilities drift in 2025.

**Recalibrate and validate the flag country by country.**

### For whom and when

Checked

## 10 of 10

Six framings stay small in every period, loan size, gender and region; rare words stay open.

**Test urgency in Asia as a hypothesis.**

**Next: the roadmap from pilot to proof**

# Start with prediction and support now; earn writing advice later

Proposed roadmap for Kiva and partners

## Now

1. Pilot the slow-funding flag at listing
2. Route lender-side support to flagged loans
3. Stop investing in framing tweaks

## Next

1. Recalibrate and validate the flag by country
2. Validate the text measures with a codebook
3. Pilot one truthful clarity prompt against a control

## Later

1. Data partnership: listing-time text, expired listings, partner identifiers
2. Answer for whom and when properly
3. Give writing advice only if the pilot shows it helps

**If a step fails, stop there: report the uncertainty and give no writing advice.**

# Thank you

**Mike Nguyen and Sophia Nguyen, Cultural Blend**

Code and evidence: github.com/tuannm3812/unsw-ma-hackathon-2026



# Regional family links look strong but rest on two countries each

Explanatory models: within-region slopes with country-clustered and heuristic few-cluster intervals




**Countries behind each region** 3

| | | | |
|---|---|---|---|
| Africa | 27 | Central America | 2 |
| Asia | 12 | Middle East | 2 |
| Oceania | 4 | North America | 1 |

**Figure A1.** Within-region family-framing slopes, 95% ranges, explanatory models; one dot per country, Kiva regions.
[1] t(1) reference on the same clustered error: a heuristic screen, not a calibrated test. Axis in log points × 100.

### 1 Both links look strong at first

Standard errors put family framing at about **7% faster funding** per extra mention in each region.

### 2 Two countries cannot carry them

A few-country check widens both ranges **across zero**; the Middle East is **97% Palestine**.

### 3 Other regions have many more countries

Africa has **27 countries** and Asia **12**; these two regions have two each.

### 4 No regional writing rules yet

Treat both as **descriptive patterns**; a real test needs more countries or partner-level data.

# Each forecast trains only on loans completed before its cutoff

Forecast benchmark: training records must have completed before each cutoff

| Evaluation year | Pre-cutoff postings completing later | Eligible as-of training | Evaluation rows |
|---|---|---|---|
| 2021 | 7,528 | 717,672 | 163,166 |
| 2022 | 3,918 | 884,448 | 143,354 |
| 2023 | 3,546 | 1,028,174 | 143,233 |
| 2024 | 5,691[1] | 1,169,262 | 141,725 |
| 2025 | 4,261 | 1,312,417 | 137,162 |

**Figure A2.** Training availability and evaluation rows by year.

[1] A posting-only split would keep these 5,691 training records; evaluation windows still select eventual completions.

# The primary models hold up on later, completed loans

Explanatory models: chronological holdout, completed loans only

**5.2031 days**

duration MAE, HistGradientBoosting

**0.5420**

duration $R^2$

**0.8997**

24-hour ROC AUC

**0.8301**

24-hour average precision

**0.1156**

24-hour Brier score

**Scope:** 1,174,953 training and 278,887 holdout loans (2024–2025). Retrospective predictions on completed loans, not deployment estimates.

# Story themes track loan purpose more than writing style

Exploratory NMF themes from a 20,000-loan sample; descriptive, not full-data evidence

## 1.5 days

mean funding time, fastest theme (sanitation)

## 13.5 days

mean funding time, slowest theme (group solar and farm plots)

### What the themes show

8 themes, including sanitation, clean water, livestock, smallholder farming and retail. They track loan purpose as much as writing style.

### How to read them

Means are unadjusted and come from a sample, so the gap is context, not a tested effect of story content.

# Text lowers a linear model's error; a stronger tree barely changes

Forecast benchmark: linear model, structure only versus structure plus text, identical evaluation rows

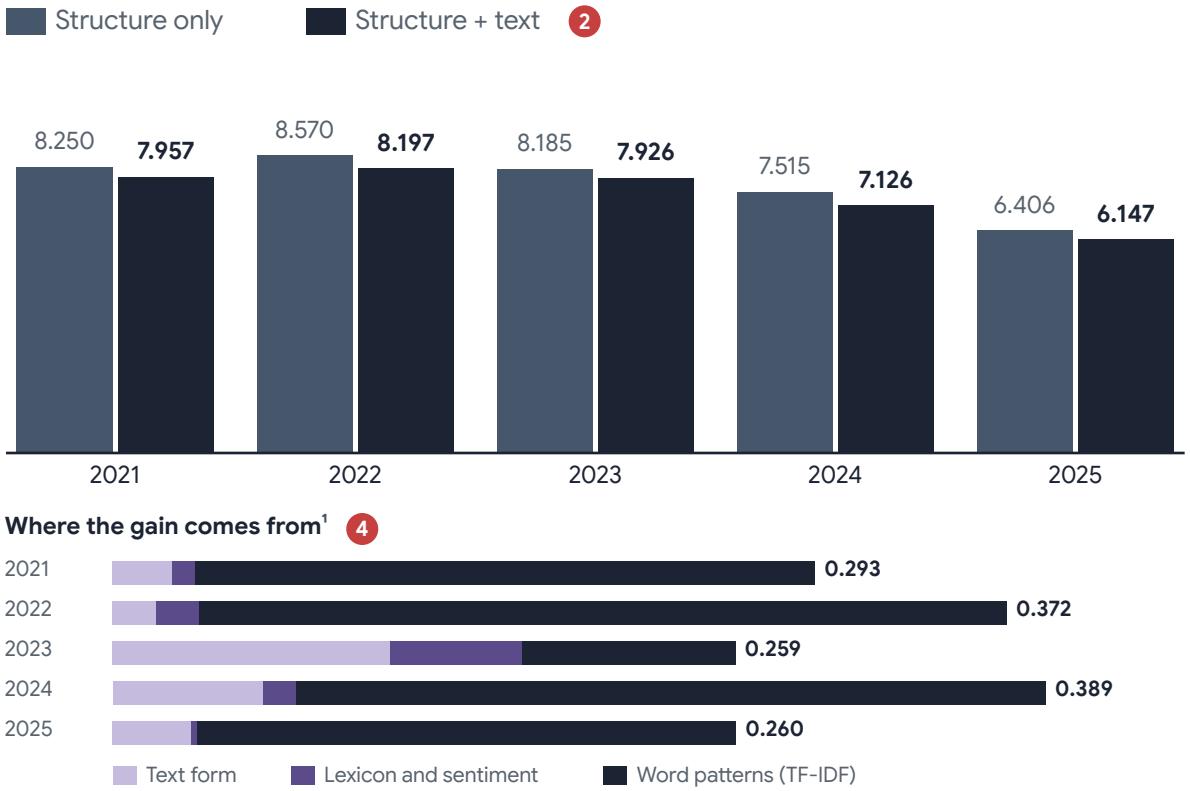



**Figure A5.** Linear-model MAE in days (top) and its reduction by text feature group (bottom), identical evaluation rows.

[1] Groups added in order: form, then lexicon and sentiment, then TF-IDF; another order would split the gain differently.

**0.29 days** 1

median annual error reduction

**5 of 5** 3

years with lower error

**728,640**

evaluation records

### 1 A small linear gain

Gains of **0.26–0.39 days** on 728,640 evaluation records; about half is context.

### 2 Forecasting is not persuasion

Text may encode **purpose, place or programme templates**, not only persuasion.

### 3 No single country drives it

Removing any single country **leaves the gain positive** in every year (exploratory check).

### 4 Word patterns carry most of it

TF-IDF gives the largest share in **4 of 5 years**; lexicon and sentiment add only **0.003–0.055 days**.

# Text became numbers through classical NLP, not a large language model

Every text feature in the deck, how it was built, and where its limits are

| Technique | What it captures | Settings | Where used | Limit |
|---|---|---|---|---|
| Description form | Length and structure of the text | Characters, words, sentences, average word and sentence length | Explanatory models and forecasts | Says nothing about content |
| Word-list counts | Eight framing measures, including first and third person | Theory-guided lists of 4–13 entries; rate per 100 words | Explanatory, robustness and equivalence; forecasts | Not yet validated (gate 1); counts words, not meaning |
| VADER sentiment | Positive or negative tone | Rule-based lexicon; compound score from –1 to 1 | Explanatory, robustness and equivalence; forecasts | General-purpose lexicon, not built for loan text |
| TF-IDF word patterns | Distinctive words and two-word phrases | Up to 5,000 features, in at least 5 training loans; fitted on training years only | Linear forecast | May pick up place and programme templates |
| Compressed text features | TF-IDF reduced for the boosted tree | Truncated SVD, 64 components; fitted on training years only | Larger boosted tree | Hard to interpret |
| NMF topics | Eight story themes | TF-IDF up to 1,000 features; 20,000-loan sample | Appendix A4 only | Exploratory; sample only, not full-data evidence |

**Figure A6.** Methods as implemented in src/ and the exploratory notebook. No large language model or pretrained embeddings were used.